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CSEN1096 Business of Software

Winter Term 2026

Course Project – Build Your (First) Startup!

For your project on this course, you will work as a founding team on a semester-long venture project. The goal of the project is to take one software startup idea **of your own choosing** through the full journey covered in this course: from an idea and a business model, through market validation, product and architecture decisions, go-to-market, growth and monetization, all the way to a pitch in front of an investor panel on Demo Day.

This is **not** a programming project, and you are not required to build a working product (you can opt to build a prototype MVP). You are required to build a convincing, evidence-based case that your startup deserves to exist and deserves to be funded. Throughout the semester your work will be evaluated the way a venture capital (VC) firm evaluates a startup: the instructor (and the panel of experts at the end) will look at the team, the problem, the market, the product, the business model, the evidence you gathered, and the numbers. At every milestone you will receive both a grade and an investment decision.

Your choice of startup idea is free, as long as it respects the following rules:

- **Software is at the core:** the main value must be delivered through software (e.g., SaaS, marketplace, consumer or mobile app, developer tool, AI product, fintech, edtech, healthtech). Hardware-enabled ideas are acceptable only if software is the core of the value and of the revenue.
- **A new venture:** the idea must be a company your team could plausibly start. It can be inspired by a company in another market ("X for Egypt"), but you must then show why it will win in your market.
- **A specific customer:** you may target Egypt, the MENA region, or a global market, but you must name a specific primary customer segment. "Everyone" is not a customer.
- **Disclosure:** if the idea builds on an existing side project, a family business, or a previous course project, declare it in the submission form.
- **Legal and ethical:** ideas that are illegal, deceptive, or designed to harm users will not be accepted.

How Your Startup Will Be Evaluated: The VC Lens

Investors do not grade ideas – they grade **investment cases**. A good idea with no evidence is a pass, and an honest team that discovers its first idea was weak will pivot and prove that the new direction is often a strong investment. The grading in this project follows the same logic: **the VC (the instructor) grades the quality of your reasoning and your evidence, not how big your dream is.** If you are unsure about your idea and would like feedback, fill in this form with a description, and the instructor will provide you with ungraded feedback through email before the end of Week 4.

The Investment Readiness Scale

Every criterion in every rubric below is rated on the same scale that a VC partner uses after a meeting:

Rating	Investor verdict	Meaning
5	Strong conviction	"I would lead this round." Clear, evidenced, and consistent with the rest of the case.
4	Interested	"Let's schedule a follow-up meeting." Strong, with minor gaps.
3	Interested, not convinced	"Come back when you have more evidence." Reasonable, but key claims are not yet supported.
2	Skeptical	"I don't see it yet." Significant gaps, weak logic, or generic content.
1	Pass	"Not for us." Claims without evidence, or the analysis is fundamentally wrong.
0	No show	The item is missing.

A criterion's score is (rating ÷ 5) × its weight. For example, a rating of 4 on a criterion weighted 15% earns 12 out of 15 points of that milestone.

The VC Scorecard

Your startup will be assessed across the following dimensions. Each milestone focuses on some of them (● primary focus, ○ revisited), so that by Demo Day the panel has seen the complete investment case.

Dimension	The question an investor asks	M1	M2	M3
Team & founder–market fit	Why are *you* the right founder(s) to build this, and how is the company structured?	●		○
Problem & customer	Is the problem real, painful, frequent and urgent? Where is the evidence?	●		○
Market opportunity	Is the market large and growing, and why is now the right time?	●		○
Product & differentiation	Is it clearly better than the alternatives, and what stops others from copying it?	●	○	○
Technical & operational feasibility	Can this be built, run and scaled at a cost that leaves a healthy margin, within the law?		●	
Go-to-market	Is there a repeatable and affordable way to reach and win customers?		●	○
Business model & unit economics	Does each customer eventually make money? Does it get better with scale?	○	○	●
Evidence & traction	What have you proven with real people, not assumptions?	●	●	●
The deal	How much do you need, what will it buy, and what do investors get?			●

After Milestones 1 and 2 you will receive a short **investment memo** from the instructor with one of three decisions: **Proceed**, **Proceed with conditions** (specific issues you must address in the next milestone), or **Pivot recommended**. The decision is feedback, not a grade: a team can receive "Pivot recommended" and still earn a high grade if its work was rigorous. Pivots after Milestone 1 are welcome. Pivots after Milestone 2 must be justified with evidence and approved by the instructor.

Project Dates

The project runs in parallel with the lectures, so each milestone uses the material covered just before it. A rough timeline of your project is:

Week	Lecture	Project activity
1	Business Vocabulary & Company Types	
2	Roles & Teams in the Product Lifecycle	Start forming teams and brainstorming ideas
3	Business Models & Monetization	Team registration due (founders, roles)
4	Market Analysis & Product-Market Fit	**Idea shortlist (one-paragraph idea, two ideas if unsure) submitted for early, ungraded feedback** Milestone 1 posted. Start customer interviews
5	Product Strategy & Prioritization	Customer discovery continues MVP scoping
6	Accounting & Finance for Software	Founding structure, equity split and vesting
7	Commercializing Your Software (GTM)	Milestone 1 deadline
8	Engineering Decisions for Business Efficiency	Milestone 1 investment memos returned Milestone 2 posted
9	Growth Loops & Retention Strategy	Validation experiment running
10	Monetizing Your Product	Milestone 2 deadline Milestone 3 posted
11	Pitch & Fundraising	Optional pitch dry-run in office hours
12	Demo Day	Milestone 3 submission and Demo Day presentations

The **Deadlines** for your project are (exact dates will be announced via the notification system and on the CMS):

- End of Week 7: Milestone 1 submission due (11:59 pm)
- End of Week 10: Milestone 2 submission due (11:59 pm)
- 48 hours before Demo Day: Milestone 3 written deliverables: financial model and final pitch deck due (11:59 pm)
- Week 12: Demo Day presentations in class – location and exact times to be announced.
- 24 hours after Demo Day: peer assessment form by classmates.

Any changes that you make to your Data Room or submitted documents **after the due date will be ignored** (the instructor will use the version history). Please have all your work submitted and checked (links, permissions, spreadsheets) before the deadline.

Project Team

You will work as a **founding team of 1-3 people**. Each founder takes one primary role (e.g., CEO, Product (CPO / product owner), Technology (CTO), Growth & Marketing (CMO), Finance & Operations (CFO/COO)) and is accountable for the matching parts of each milestone, although all founders are expected to contribute to customer discovery and to the pitch. Roles are recorded in your founders' agreement in Milestone 1 and can change later if you document why.

In general, the instructor does not anticipate that the grades for each group member will be different. However, the instructor reserves the right to assign different grades to each group member based on the contribution tables and peer assessments (see below).

Project Milestones

It is critical to note that **no extensions will be given** for any of the project submission due dates declared above. Demo Day sessions may be rescheduled in the revision week, subject to your own loads. Milestones submitted after their due date will not be graded. If you anticipate any issues (e.g., due to travel plans) you need to send the instructor an email at least one week in advance.

Each milestone corresponds to a stage of a real fundraising process. Every milestone submission must include a one-page **Founders' Contribution Table** (template uploaded to CMS, and is available with working example here) stating who did what, and every milestone after the first must start with a one-page **Changelog** explaining what changed since the previous milestone.

Milestone 1: Project Idea & Business Model Canvas

Posted Week 4 · Due end of Week 7 · VC stage: screening and first meeting

In a first meeting, an investor wants to know whether there is a real problem, a real customer, a big enough market, and a team that understands all three. This milestone builds that case.

Deliverable 1.1: Founding Memo (max. 12 pages, excluding appendices)

- **Vision and mission:** where the company is going in 10 years and what it does every day to get there.
- **Problem and customer:** who has the problem, how often, how they solve it today, and what the problem costs them. Define your **Ideal Customer Profile (ICP)** and write your core **Job-to-be-Done** statement.
- **Customer discovery:** at least **5-7 interviews** with people who match your ICP (distributed over founders), run using *The Mom Test* principles: **ask about past behavior, not opinions about your idea**. Include your interview guide, a summary of findings, and what surprised you or invalidated your assumptions. The full interview log goes in an appendix (anonymized, with date, role/segment and interviewer).
- **Market sizing: TAM, SAM and SOM**, calculated both top-down and bottom-up, with every assumption and source stated, and a short "why now?" argument.
- **Market structure and competition:** direct competitors, indirect competitors and substitutes, including the "do nothing / spreadsheet" option. Present a competition matrix[1] and a positioning map[2].
- **Differentiation and moat:** what makes you clearly better today, and which moat (e.g., network effects, proprietary data, switching costs, brand, scale economics) protects you tomorrow.
- **Product-market fit plan:** how you will know you have reached PMF (e.g., Sean Ellis 40% test, Superhuman-style segmentation, retention signals) and the target thresholds.
- **Product strategy and MVP:** MVP scope designed to test your riskiest assumption, a prioritized feature list using **MoSCoW** (Must-haves, Should-haves, Could-haves, Won't-haves) and a **Kano classification** (Basics, one-dimensionals, delighters, indifferent) of your top 15-20 features, and a Now / Next / Later roadmap for the first 6–12 months.
- **Company type and founding structure:** is this a venture-scale startup or an SME/lifestyle business, and why? Founding roles (including who owns product), proposed equity split with

[1] Example templates: https://www.smartsheet.com/free-competitor-analysis-template?srsltid=AU7gw4VSDnTegcrNiIaXdr_SVEoLBt4B-OSGtyKL61v0Oed24YKZ93BX

[2] Example template: https://www.canva.com/templates/EAGzAAAjirs-beige-minimal-brand-positioning-map-whiteboard/

vesting (e.g., 4 years with a 1-year cliff), and IP assignment to the company. Use the Cap table template with explanation here for reference. Template is available on the CMS.

Deliverable 1.2: Business Model Canvas and Pricing Hypothesis

A one-page Business Model Canvas (BMC) with all nine blocks, consistent with your interview findings. Add a short pricing hypothesis: which monetization model you will use (e.g., SaaS tiers, freemium, marketplace take rate, usage-based), the initial price point, and the reasoning behind it (value-based pricing, anchoring, customers' willingness to pay from interviews).

Deliverable 1.3: Riskiest Assumptions

The five assumptions that would kill the company if they were wrong, ranked by risk, each with the cheapest test you could run to check it. One of them will become your validation experiment in Milestone 2.

Deliverable 1.4: Investor One-Pager

A single-page teaser you would email to an investor to get a meeting: problem, solution, customer, market, model, evidence so far, team.

Milestone 1 Rubric

Criterion (weight)	5 – Strong conviction	3 – Interested, not convinced	1 – Pass
Problem & Customer Evidence (25%)	• 10+ well-run interviews with the ICP • Questions about past behavior • Pain, frequency and current workarounds are clear • Findings changed at least one assumption	• Interviews done but partly leading, or with friends and non-target users • Pain is plausible but intensity is unclear	• Few or no real interviews • Problem is asserted rather than evidenced
Market Opportunity (15%)	• Top-down and bottom-up sizing reconcile • Sources and assumptions explicit • Credible "why now"	• Numbers present but mostly top-down or weakly sourced • SOM not tied to a realistic capture plan	• One large "1% of a huge market" number with no sources
Competition & Differentiation (15%)	• Direct, indirect and do-nothing alternatives mapped • Specific, defensible moat that strengthens over time	• Competitors listed, but differentiation is feature-level and easy to copy	• Claims "no competitors" • No moat
Business Model Canvas & Pricing (15%)	• All nine blocks consistent with each other and with the evidence • Monetization model fits the customer • Value-based price hypothesis	• Canvas complete but generic • Price is cost-plus or guessed	• Blocks missing or contradictory • No revenue logic

Criterion (weight)	5 – Strong conviction	3 – Interested, not convinced	1 – Pass
Product Strategy & MVP (15%)	• Tight MVP that tests the riskiest assumption • Prioritization framework applied transparently • Realistic roadmap	• MVP too large or not linked to assumptions • Framework applied superficially	• Feature wish-list with no prioritization
Team & Founding Structure (5%)	• Clear roles and ownership • Founder–market fit argued • Equity split with vesting and IP assignment justified	• Roles listed with thin rationale • Equity split without vesting	• No roles or structure
Communication & One-Pager (10%)	• Concise, well-structured and persuasive • One-pager that makes an investor want a meeting	• Complete but wordy or unfocused	• Hard to follow or exceeds page limits

Submission Instructions

Milestone reports are submitted as a link to a folder containing the relevant PDF files on a form by the founder or one of the co-founder(s). In addition, every team keeps a single shared **Data Room** (a Google Drive or OneDrive folder) whose link should be provided in the form as well. All links should have view access for the instructor. For the BMC, either use the PPTX template provided on the CMS, or use any online template you see fit from Canva. **If the instructor cannot access your work because these directions are not followed correctly, your work will not be graded.** Over time, the Data Room must contain:

- *Milestone1, Milestone2, Milestone3* – the submitted reports and their appendices.
- *Evidence* – the interview log, experiment results and any LOIs, pre-orders or analytics exports.
- *Financials* – the financial model (with working formulas, no pasted values) and the cap table.
- *Legal* – the founders' agreement (roles, equity, vesting, IP assignment).
- *Pitch* – the final pitch deck (PDF) and, optionally, a demo video or prototype link.
- *README* – a one-page index of the Data Room with the team name, founders, roles, and links to every item.

For Milestone 1, the Data Room should include the **Founders' Contribution Table** (one page)[3]**,** the Founding Memo, the cap table, the BMC, riskiest assumptions document, Investor One-Pager, and all evidence related to verifying the idea is worth building.

Milestones submitted after their due date will not be graded. Late days do not apply. After Demo Day, you will fill out a peer evaluation, which will be published on Google Forms right after the last Demo Day session.

[3] If you want to use the comprehensive Founder Agreement, a template can be found here: https://promise.legal/templates/founder-agreement

Academic Integrity and Use of AI

You may use generative AI tools for research, brainstorming, drafting and building spreadsheets, provided you state in an appendix of each milestone which tools you used and for what. You remain responsible for every claim and number you submit, and all market data must be cited to its original source. The idea seed should be real and not proposed by AI.

Customer interviews, experiment results and any other evidence **must come from real people and real activity**. Fabricated or AI-generated interview data or results are academic misconduct and will result in a zero for the whole project. The instructor may ask to see raw evidence or, with the interviewee's consent, contact them to confirm an interview took place.

Grading Criteria

The project is worth 40% of your total course grade. These 40% will be divided as follows:

Component	**Weight**	**Graded with**
Milestone 1: Project Idea & Business Model Canvas	25%	Milestone 1 Rubric
Milestone 2: Software Architecture & Marketing Plan	25%	Milestone 2 Rubric
Milestone 3: Growth & Monetization Plans (written deliverables)	20%	Milestone 3 Rubric
Demo Day Pitch	30%	Demo Day Rubric
Total	**100%**	
Best Investment bonus	*up to +5%*	Investor panel allocation

Across all milestones, the grade corresponds to answers to questions that a VC would ask: *Did you choose a problem worth solving? Did you gather real evidence? Is your reasoning sound and consistent? Would a rational investor put money into this company?*